模电作业.10

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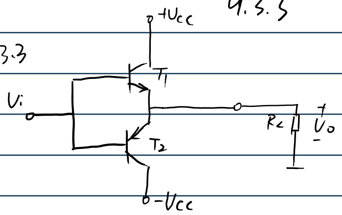
2025年12月23日 21:24

9.3.3 9.4.5

9.3.3



(1) $P_{om} = \frac{1}{2}\frac{V_{CC}^2}{R_L}$

$V_{CC}=12V$

(2) $I_{CM} \geq \frac{V_{CC}}{R_L} = \frac{12}{8} = 1.5A$.

$|V_{(BR)CEO}| \geq 2V_{CC} = 24V$.

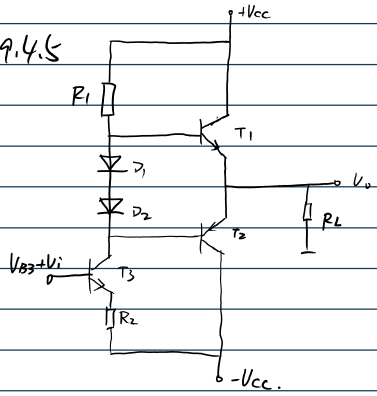
(3) $P_V = \frac{2V_{CC}^2}{\pi R_L} = \frac{2\times 12^2}{8\pi} = \frac{36}{\pi}$

(4) P_{CM}.

$P_{CM} \geq 0.2 P_{om} = 1.8W$.

(5) $V_i \approx V_o \approx \frac{1}{\sqrt{2}} V_{CC} = 6\sqrt{2}V$.

9.4.5



P_o. P_V. P_T. η.

$P_o = \frac{V_o^2}{R_L} = \frac{16^2}{16} = 16W$

$P_T = \frac{2}{R_L}\cdot\left(\frac{V_{CC}V_{om}}{\pi} - \frac{V_{om}^2}{4}\right) = 5.6W$.

∴ $P_V = P_o + P_T = 21.6W$.

∴ $\eta = \frac{P_o}{P_V} = \frac{16}{21.6}\times 100\% = 74.1\%$